Mathematical Foundations of Machine Learning

Problem 1 (25 points) Consider the following system of linear equation for some constants b and g:

$$1x + by + 0z = 1$$
$$1x - 2y - 1z = 2$$
$$0x + 1y + 1z = g$$

a) Which number b leads to a row exchange?

b) Which number b makes the system singular (missing a third pivot)?

c) For the b found in part (b), for which value of g, the system has infinitely many solutions?

d)Find the solutions of the system with b and g found in parts (b) and (c), if z = 2.

a) $\begin{bmatrix} 1 & b & 0 \\ 1 & -2 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ g \end{bmatrix} \xrightarrow{R2=R2-R1} \begin{bmatrix} 1 & b & 0 \\ 0 & -2-b & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ g \end{bmatrix}$

We will have a row exchange if $-2-b=0 \Rightarrow b=-2$.

b) R2 to be a multiple of R3, $-2-b=-1 \Rightarrow b=-1$

c) $\begin{bmatrix} 1 & -1 & 0 \\ 1 & -2 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ g \end{bmatrix} \xrightarrow{R2=R2-R1} \begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & -1 \\ 0 & 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ g \end{bmatrix}$

$\xrightarrow{R3=R3+R2} \begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ g+1 \end{bmatrix}$ $\quad g+1=0 \rightarrow g=-1$

$0x+0y+0z=g+1$

$g=-1$ system has an infinite nr of sols.

d) $\begin{bmatrix} 1 & -1 & 0 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ $\quad z=2$ $\quad -y-z=1$ $\quad y=-3$ $\quad x-y=1$ $\quad x+3=1$ $\quad x=-2$

$Sol = \begin{bmatrix} -2 \\ -3 \\ 2 \end{bmatrix}$

Problem 2 (25 points) Find a basis for each of the four fundamental subspaces of the matrix $A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ 0 & 0 & 2 & 4 \end{bmatrix}$.

$\begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ 0 & 0 & 2 & 4 \end{bmatrix} \xrightarrow[R3=R3+R2]{R2=R2-2R1} \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R1=R1-R2} \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ RREF

a) $C(A) = \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} \right\}$ basis for $C(A)$

b) $\begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

b, d are free var

$b=1, d=0 \rightarrow c+2d=0 \Rightarrow c=0, \; a+3b=0 \rightarrow a=-3$ $\begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}$

$b=0, d=1 \rightarrow c+2d=0 \rightarrow c=-2, \; a+3b=0 \rightarrow a=0$ $\begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix}$

Basis $N(A) = \left\{ \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$

c) $C(A^T)$

Basis $C(A^T) = \left\{ \begin{bmatrix} 1 \\ 3 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ 4 \\ 8 \end{bmatrix} \right\}$

d) $N(A^T)$

$A^T = \begin{bmatrix} 1 & 2 & 0 \\ 3 & 6 & 0 \\ 1 & 4 & 2 \\ 2 & 8 & 4 \end{bmatrix}$ Elimination $\sim$ $A^T x = 0 \rightarrow \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}$

Basis $N(A^T) = \left\{ \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \right\}$

Problem 3 (25 points) Suppose the complete solution to the equation $Ax = \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix}$ is

$$x = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}.$$

x, b

(a) What is the dimension of the row space of A? 1

(b) What is the matrix A?

(c) Describe exactly all the vectors b for which $Ax = b$ can be solved. (Don't just say that b must be in the columns space of A.)

$\begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}$ is particular sol, $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$ are in $N(A)$

$$\begin{bmatrix} a & d & g \\ b & e & h \\ c & f & i \end{bmatrix} \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix} \qquad \begin{array}{l} 2a = 2 \to a = 1 \\ 2b = 4 \to b = 2 \\ 2c = 4 \to c = 2 \end{array}$$

$$\begin{bmatrix} 1 & d & g \\ 2 & e & h \\ 2 & f & i \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad \begin{array}{l} 1+d = 0 \to d = -1 \\ 2+e = 0 \to e = -2 \\ 2+f = 0 \to f = -2 \end{array}$$

$$\begin{bmatrix} 1 & -1 & g \\ 2 & -2 & h \\ 2 & -2 & i \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \qquad \begin{array}{l} g = 0 \\ h = 0 \\ i = 0 \end{array}$$

Rank = 1

$$A = \begin{bmatrix} 1 & -1 & 0 \\ 2 & -2 & 0 \\ 2 & -2 & 0 \end{bmatrix}$$

c) $x = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} + c \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} + d \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 2+c \\ c \\ d \end{bmatrix}, \forall c, d \in \mathbb{R}.$

Problem 4 (25 points)

(a) Find an orthonormal basis for the subspace S of $\mathbf{R}^4$ spanned by these three vectors:

$$a_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, a_2 = \begin{bmatrix} 3 \\ 1 \\ 1 \\ 3 \end{bmatrix}, a_3 = a_1 + a_2$$

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(b) Find the closest vector p in that subspace S to the vector $b = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix}$.

a) Basis $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 3 \\ 1 \\ 1 \\ 3 \end{bmatrix} \right\}$ Gram Schmidt

$q_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}$ unit $\begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}$

$B = q_2 - \frac{q_1^T q_2}{q_1^T q_1} q_1 = \begin{bmatrix} 1 \\ -1 \\ -1 \\ 1 \end{bmatrix}$ unit $\begin{bmatrix} 1/2 \\ -1/2 \\ -1/2 \\ 1/2 \end{bmatrix}$

$\left\{ \begin{bmatrix} 1/2 \\ 1/2 \\ 1/2 \\ 1/2 \end{bmatrix}, \begin{bmatrix} 1/2 \\ -1/2 \\ -1/2 \\ 1/2 \end{bmatrix} \right\}$

$q_3 = q_1 + q_2$ does not affect the basis

b) $p = QQ^T b = \begin{bmatrix} 1/2 \\ 0 \\ 0 \\ 1/2 \end{bmatrix}$